Elbert Chao

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Education

Western University Sept. 2025 – Present
Master of Engineering Science (MESc), Software Engineering *London, ON*
Thesis: Uncertainty Quantification for Deep Learning in Medical Imaging
Award: Med-i CREATE Competitive Scholarship ($10,000)
Coursework: Digital Image Processing, Dependable AI Systems, LLM/VLM, Data Analytics

Bachelor of Engineering Science (BESc), Software Engineering Sept. 2021 – May 2025
London, ON
Honours: Dean's Honours List (4×) — **Year 4 GPA: 3.9/4.0**
Coursework: Machine Learning, Artificial Intelligence I/II, Data Structures & Algorithms, Cloud Computing

Technical Skills

Programming Languages: Python, TypeScript
ML & AI: PyTorch, TensorFlow Lite, OpenAI API, LangChain, Prompt Engineering, Computer Vision, CNN, Transformer
Backend & Data: FastAPI, Flask, MongoDB, REST APIs **Tools**: GCP, Streamlit, Nginx, Jira, Confluence

Experience

PM Accelerator Jun. 2025 – Aug. 2025
AI Engineer Intern (Python, FastAPI, OpenAI API, MongoDB) Remote (Boston, MA)

- Built a semantic resume to job matching system using **OpenAI API** and **FastAPI**, improving alignment accuracy by **35%.**
- Developed a Python web scraping and text normalization pipeline to ingest unstructured job data, reducing manual processing by **60%.**
- Designed a scalable **RESTful API** to support resume parsing and context-aware interview question generation.

Free Appropriate Sustainability Technology – FAST Nov. 2024 – Apr. 2025
Full-Stack Developer (Next, Python, Flask, Nginx, PM2) London, ON

- Developed a full-stack solution with a **Next frontend (TypeScript)** and **Flask backend (Python)** to deliver ML-based insights into strawberry health, resulting in a **90%** increase in crop monitoring effectiveness.
- Enhanced AI model performance by conducting real-world tests, systematically debugging edge cases and refining feature inputs, achieving classification speeds under **500ms**.
- Maintained server uptime across a **2-month** period with **fewer than 3** total downtime incidents using Nginx and PM2 for load balancing and process management.
- Configured and secured a production environment for a campus-deployed web service, enabling **24/7** availability throughout the term with minimal downtime or configuration drift.

Projects

Probabilistic Deep Learning for Kidney Ablation Support Aug. 2025 – Present
Graduate Researcher (Python, PyTorch, CNNs, Transformers)

- Refined a **hybrid CNN–Transformer** segmentation model with custom stratified resampling to address severe data scarcity, improving Dice score by **27%** (58% → 85%).
- Implemented **Monte Carlo Dropout** and **layer-wise uncertainty heatmaps** to quantify epistemic uncertainty, producing pixel-level probability maps to flag low-confidence regions for clinical review.
- Built a scalable validation pipeline over **1,500 CT scans (125 patients)** to measure prediction variance and quantifying model reliability in the presence of anomalies.

Warren Buffett Chatbot Agent Jun. 2025 – Aug. 2025
AI Engineer (Python, OpenAI API, LangChain)

- Built an end-to-end conversational AI system using **LangChain** and **OpenAI models**, achieving **95%** response accuracy in simulating Warren Buffett–style investment reasoning.
- Integrated external financial and search APIs to support real-time retrieval and multi-tool reasoning, enabling robust handling of **100+** distinct query types.

FormFixer.AI — 4th Year Capstone Project Sept. 2024 – Apr. 2025
Team Lead (TensorFlow Lite, Computer Vision)

- Optimized a **TensorFlow Lite** pose estimation model for real-time inference, reducing joint misclassification rates by **80%** during live video analysis.